

The Economics of Bicycles for the Mind

Cognitive tools, human judgment, and inequality

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Agrawal, Gans & Goldfarb (2025) · NBER WP 34034
github.com/gsaco/ai-02-agrawal

The paper and the agent's problem

Question and single mechanism

Question. When a cognitive tool makes implementation easier, how do effort, productivity, returns to human abilities, and inequality change?

Mechanism. The tool is a **substitute for implementation effort** at the margin; payoff judgment α and opportunity judgment $\gamma(t)$ determine how much value the agent can obtain from it.

Agent's problem, conditional on an opportunity

Choose effort e_t to maximize expected improvement value net of cost:

$$e_t^*(\theta) \in \arg \max_{e_t \geq 0} \{p(se_t; \theta)\alpha\Delta - c(e_t; \theta)\}.$$

Interior FOC: $p'(se_t^*; \theta)s\alpha\Delta = c'(e_t^*; \theta)$.

Choice and constraint: $e_t \geq 0$.

Technology: $p \in [0, 1]$ is increasing and weakly concave.

Cost: c is increasing and weakly convex.

Tool: raises p and/or lowers c , while lowering p'/c' as θ rises.

Propositions 1–2: effort, value, and adoption

Proposition 1 · More value with less effort

As tool quality moves from $\theta = 0$ to $\theta = 1$,

$$e_t^*(1) < e_t^*(0) \quad \forall t, \quad e_t^*(\theta) = e^*(\theta) \quad \forall t, \\ V_0(1) > V_0(0).$$

Intuition. The same per-opportunity problem is solved each period; the tool lowers optimal effort while raising optimized task value.

Definition 1's direct effects and lower p'/c' apply, with a well-defined interior optimum.

Proposition 2 · What drives adoption value?

$$\Gamma = \sum_{t=0}^{\infty} \left(\prod_{i=0}^t \gamma(i) \right) \delta^t, \quad \Delta V = \Gamma [M_1^* - M_0^*] > 0.$$

Opportunity judgment. Γ multiplies the entire gain.

Payoff judgment.

$$\frac{\partial \Delta V}{\partial \alpha} = \Gamma \Delta [p(se^*(1); 1) - p(se^*(0); 0)].$$

Complementarity requires realized success to be weakly higher with the tool.

Implementation skill. Adoption value falls with s under the tool–skill substitutability condition $p_{s\theta} < 0$.

Timing. Earlier opportunities receive greater discounted weight.

Proposition 3: the variance result

Specialized model and exact result

Setup

$p(se; \theta) = \sqrt{se + \theta}$, $c(e) = e$, $\theta \geq 0$.

$\gamma(0) = \gamma_0$, $\gamma(t) = \gamma$ for $t > 0$; $\delta\gamma < 1$.

$\alpha, \gamma_0, \gamma, s$ independent, positive support;
 $\mu_i > 3\sigma_i$.

Interior range:

$$e^*(\theta) = \frac{\alpha^2 \Delta^2 s}{4} - \frac{\theta}{s} > 0.$$

Value

$$\Gamma = \frac{\gamma_0}{1 - \delta\gamma},$$

$$V(\theta) = \Gamma \left(\frac{\alpha^2 \Delta^2 s}{4} + \frac{\theta}{s} \right).$$

Object: cross-sectional wage variance
 $\text{Var}[V(\theta)]$ as continuous tool quality θ
changes.

Conditional U-shape

$$\frac{\mathbb{E}[\Gamma^2]}{\mathbb{E}[\Gamma]^2} < \mu_s \mathbb{E}[1/s] \quad (30)$$

with $\text{Var}(\Gamma/s) > 0$ and an interior turn.
Define

$$a_0 = \frac{\Delta^2 \mathbb{E}[\alpha^2]}{2} \times \left(\mathbb{E}[\Gamma^2] - \mathbb{E}[\Gamma]^2 \times \mu_s \mathbb{E}[1/s] \right) < 0,$$

$$\frac{d \text{Var}[V(\theta)]}{d\theta} = a_0 + 2\theta \text{Var}(\Gamma/s),$$

$$\theta^* = -\frac{a_0}{2 \text{Var}(\Gamma/s)} > 0.$$

Variance audit

What is U-shaped?

$\text{Var}[V(\theta)]$ across agents as θ changes.

Intuition: inverse skill bias equalizes first;
heterogeneous Γ/s eventually widens
wages.

Internal slip

Condition (30) gives the initial fall. A
positive interior slope at $\theta = 1$ also
requires $\theta^* < 1$.

Individual benefit is different

$D_i(\theta) = \theta \Gamma_i / s_i$, so

$$\text{Var}[D(\theta)] = \theta^2 \text{Var}(\Gamma/s),$$

which rises monotonically when Γ/s is
heterogeneous.